F78AA1 DECEMBER 2014

SOLUTIONS

1. Let $t=0$ be 1/1/15 and choose the year as the time unit.

(a) $PV = \sum_{r=1}^{5} C_r V(0,t_r)$

$= 100\,V(0,1/2)$
$+ 100\,V(0,1\tfrac{1}{2})$
$- 25\,V(0,2\tfrac{3}{4})$
$- 50\,V(0,3\tfrac{1}{2})$
$- 75\,V(0,4\tfrac{1}{2})$ [2]

$= 100\,(1.05)^{-1/2}$
$+ 100\,(1.05)^{-1}(1.025)^{-2}$
$- 25\,(1.05)^{-1}(1.025)^{-7}$
$- 50\,(1.05)^{-1}(1.025)^{-8}(1.01)^{-6}$
$- 75\,(1.05)^{-1}(1.025)^{-8}(1.01)^{-18}$ [2]

$= 97.59 + 90.65 - 20.03 - 36.82 - 49.01$

$= £82.38$ [1]

(b) Accumulation $= 82.38\,(1.05)^{1}(1.025)^{8}(1.01)^{36}$ [1]

$= £150.79$ [1]

[Total 7]

2. During the first 5 years we have

$$PV[\text{Payments } 1\text{–}5] = 1000\left[1 + \frac{1.04}{1.07} + \ldots + \frac{1.04^4}{1.07^4}\right] \qquad [1]$$

$$= 1000\, \ddot{a}^{j}_{\overline{5}|} \quad \text{where } j = \frac{1.07}{1.04} - 1 = 0.028846 \qquad [1]$$

$$= £4,727.38$$

$$PV \text{ at } t=5\,[\text{Payments } 6\text{–}15] = 1000(1.04)^5\left[1 + \frac{1.04}{1.1} + \ldots + \frac{1.04^9}{1.1^9}\right] \qquad [1]+[1]$$

$$= 1000(1.04)^5\, \ddot{a}^{j'}_{\overline{10}|} \quad \text{where } j' = \frac{1.1}{1.04} - 1 = 0.057692$$

$$= £9,575.72 \qquad [1]$$

Hence p.v. of all payments, at $t=0$, is

$$4,727.38 + (1.07)^{-5} \times 9,575.72 = £11,554.73 \qquad [1]$$

[Total 6]

3 (a) (i) $P = \dfrac{200,000}{a_{\overline{30}|}^{0.08}} = \dfrac{200,000}{\left(\dfrac{1-1.08^{-30}}{0.08}\right)} = £17,768.49$ [2]

(ii) Cap o/s $= P a_{\overline{20}|}^{0.08} = P\left(\dfrac{1-1.08^{-20}}{0.08}\right) = £174,474.02$ [1]

(iii) Int cost = Cap o/s $\times 0.08 = £13,957.92$ [1]

(b) After three years the Cap o/s has increased to

$174,474.02(1.1)^3 = £232,158.68$ [1]

The remaining term is then 17 years and the new repayment is the solution of:

$P'\left(a_{\overline{3}|}^{0.1} + 1.1^{-3} a_{\overline{14}|}^{0.08}\right) = 232,158.68$ [2]

ie $P(2.486852 + 6.194017) = £26,748.71$ [2] + [1]

[Total 10]

4. (a) $\frac{g(1-t_1)}{R} = \frac{0.05(1-0.25)}{1-0.2} = 0.036765 < 0.044031$ [1]

↑ Effective yield per half-year.

therefore CGT will be payable.

$$P = 5 \times 0.75 \times a_{\overline{40}|}^{0.044031} + \left(102 - 0.15(102-P)\right)(1.09)^{-20}$$ [2]

↑ $= 18.658933$; ↑ 0.178431

$= 69.971000$

$= 69.971000 + 15.469968 + 0.026765\,P$ [1]

$\Rightarrow P = \frac{69.971000 + 15.469968}{1 - 0.026765} = £87.79.$ [1].

(b) Net coupon per half-year $= 5 \times 0.75 = 3.75$ [1]

$\Rightarrow$ Net coupon per £100 of purchase price $= \frac{3.75}{0.8779}$ [1]

$= 4.271557.$

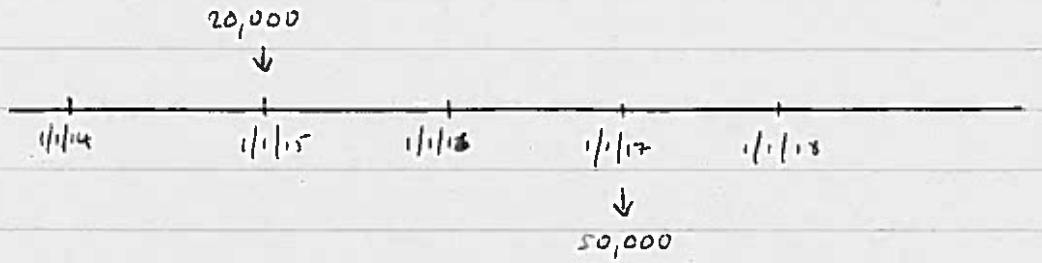
$\Rightarrow$ Effective net yield per half-year $= 0.04271557$ [1]

$\Rightarrow$ Effective net yield p.a. $= 0.087256$ or 8.72% [1].

(c) Lower yield, because the investor would not receive the interest accrued during the last 1/4 year. [1]

[Total 10]

5.



100,000	102,000	130,000	129,000	90,000	FUND

(a) 2014-15 : $100(1+i_1)^2 + 20(1+i_1) - 130 = 0$ [1]

$$\Rightarrow (1+i_1) = \frac{-20 \pm \sqrt{20^2 + 4 \cdot 130 \cdot 100}}{200}$$

$= 1.044552$ (other solution is negative) [1]

2016-17 : $130(1+i_2)^2 - 50(1+i_2) - 90 = 0$ [1]

$$\Rightarrow (1+i_2) = \frac{50 \pm \sqrt{50^2 + 4 \cdot 130 \cdot 90}}{260}$$

$= 1.046292$ (other solution is negative) [1]

Hence $LIRR = \left((1+i_1)^2(1+i_2)^2\right)^{1/4} - 1 = 0.045422$ [1]

(b) $(1+TWRR)^4 = \frac{102}{100} \cdot \frac{170}{122} \cdot \frac{129}{130} \cdot \frac{90}{79}$ [1]

$\Rightarrow TWRR = 0.052838.$ [1]

[Total 7]

6 (a) Principle of Equivalence : If two set of cashflows have the same value at any point of time, under given interest rates, we are indifferent which set we possess, or would be willing to exchange one for the other [1]

It is used as a criterion to establish a preference for holding an investment where cashflows have higher value. [1]

(b) A: Yield $= (5/2)^{1/10} - 1 = 0.095958$ p.a. [1]

B: Yield $= \frac{2200}{20,000} = 0.11$ p.a. [2]

C: Equation of value at $t = 10$

$$f(i) = 20,000(1+i)^{10} - 3,600\, s_{\overline{10}|}^{\,i} = 0 \qquad [2]$$

$f(0.1) = -5,499.9$
$f(0.11) = -3,410.8$
$f(0.12) = -1,058.5$ [2]
$f(0.13) = +1,580.3$

linear interpolation

$$i^* \cong 0.12 + 0.01 \frac{1058.5}{1058.5 + 1580.3} \qquad [1]$$

$$= 0.124$$

Hence we would prefer C.

[Total 10]

7 (a) 6.1678% p.a. is equivalent to 0.5% per month.

We need to find the smallest t such that

$$1{,}000(1+i)^t \leq 25\, s_{\overline{t}|}^{0.005} \qquad [1]$$

ie $$1000(1.005)^t \leq 25\left(\frac{1.005^t - 1}{0.005}\right)$$

ie $$\left(1{,}000 - \frac{25}{0.005}\right)(1.005)^t \leq \frac{-25}{0.005}$$

ie. $$-4000(1.005)^t \leq -5{,}000$$

[3]

ie $$(1.005)^t \geq 1.25$$

ie $$t \geq \frac{\log 1.25}{\log 1.005}$$

(or any equivalent steps)

ie $$t \geq 44.74$$

So the DPP is 45 months. [1]

(b) 4.9070% p.a. is equivalent to 0.4% per month

Profit after 45 months is (thousands)

$$25\, s_{\overline{45}|}^{0.005} - 1{,}000(1.005)^{45} = 6.483 \qquad [2]$$

Hence accumulated profit after 60 months (thousands) is

$$6.483(1.004)^{15} + 25\, s_{\overline{15}|}^{0.004} + 7{,}000 - \frac{5{,}000}{(1.004)^{60}} \qquad [2]$$

$= 6.883 + 385.684 + 7000 - 6853.20$ (thousands)

$= £1,039,167$ [1]

[Total 10]